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A user-centred approach to enhancing engagement and task management for ADHD students using AR and generative AI

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Abstract

This pilot study investigated the potential of integrating Augmented Reality (AR) and Generative Artificial Intelligence (GenAI) to support engagement and task management for university students diagnosed with Attention Deficit Hyperactivity Disorder (ADHD). Using a Design-Based Research (DBR) approach, we evaluated ADHDvance LearnAR across three within-subject conditions (baseline tools, AR-enhanced, and AR+GenAI) with $n = 15$ undergraduate participants. Quantitative measures showed directional improvements in engagement duration, interaction frequency, and task completion across conditions; however, differences did not reach statistical significance, consistent with the study's exploratory scale. Qualitative analysis provided explanatory depth, identifying themes related to (i) increased task salience through spatial AR visualisation, (ii) improved task initiation and planning support, and (iii) reduced perceived cognitive load through just-in-time GenAI scaffolding. Findings suggest AR+GenAI may offer promising affordances for inclusive learning design, warranting longitudinal evaluation with larger samples and stronger causal designs.

Keywords Augmented Reality in Education, Generative AI in Learning, Adaptive

1 Introduction

Attention-Deficit Hyperactivity Disorder (ADHD) is a prevalent neurodevelopmental disorder marked by chronic patterns of inattention, impulsivity, and hyperactivity, significantly impacting individuals across educational, social, and occupational domains.

Attention-Deficit Hyperactivity Disorder (ADHD) is a neurodevelopmental condition characterised by persistent inattention, impulsivity, and hyperactivity, affecting educational and social functioning. Globally, ADHD impacts 5–10% of children, with many continuing to experience symptoms into adulthood [1]. In higher education, students with ADHD face particular challenges in time management, task organisation, sustained attention, and self-regulation, skills essential for academic success [2].

While pharmacological treatments may alleviate core symptoms, they often fall short in addressing executive function difficulties such as task initiation, planning,



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and attentional control [3, 4]. These deficits are exacerbated in university contexts that demand autonomous learning and effective personal management. Traditional strategies, such as lectures and static task planners, frequently fail to support these learners, potentially increasing dropout rates and academic stress.

In response, emerging technologies offer new pathways for support. Augmented Reality (AR) creates immersive and interactive environments that can foster focus and reduce cognitive overload [5], while Generative AI (GenAI) enables adaptive, personalised learning experiences responsive to learners' needs [4]. Together, these technologies hold promise for addressing the fluctuating attention, impulsivity, and executive dysfunction associated with ADHD in higher education.

Recognising the potential of these technologies to improve educational outcomes for students with ADHD, this study introduces *ADHDvance LearnAR*—a novel educational application that integrates AR and GenAI to provide user-centred, adaptive support. Prior studies have demonstrated that AR enhances attention, memory, and engagement in students with ADHD by providing multimodal, interactive learning experiences [6, 7]. Similarly, GenAI-driven interventions such as AI tutors or chatbots have shown promise in supporting reading comprehension, organisation, and metacognitive development through personalised, real-time feedback [8, 9]. However, these technologies are rarely combined or specifically tailored to address the executive function challenges faced by ADHD students in higher education. By uniting the visual immediacy of AR with the adaptive responsiveness of GenAI, this study aims to explore a complementary, hybrid approach to reducing cognitive overload, improving task management, and fostering inclusive learning environments for neurodiverse learners.

The development of *ADHDvance LearnAR* adopts a rigorous Design-Based Research (DBR) approach, an iterative methodology particularly suitable for educational technology research. DBR emphasises collaboration with end-users (students and educators), iterative design cycles, and mixed-methods evaluation to systematically refine and enhance educational interventions. By engaging directly with ADHD students throughout the application's development process, the research ensures that *ADHDvance LearnAR* effectively meets its target users' practical and cognitive requirements.

This study specifically addresses the following research questions:

- How do AR visualisations and interactive features influence engagement and task completion among university students with ADHD?
- What effects do Generative AI-driven personalised learning experiences have on the organisational skills and academic performance of students with ADHD?
- Can the integration of AR and GenAI technologies effectively reduce cognitive overload and manage behavioural symptoms such as impulsivity and inattention in students with ADHD?
- What implications do the combined use of AR and GenAI technologies have on students' long-term academic and social outcomes?

Through exploring these research questions, this study aims to significantly contribute both theoretically and practically to special education contexts, particularly regarding ADHD. The study's findings are expected to provide meaningful insights into the potential of integrated AR and GenAI technologies to enhance engagement, organisational skills, and academic success among undergraduate students with ADHD. By doing so,

this research not only advances knowledge in the domain of technology-enhanced learning but also informs the development of future inclusive educational tools, underscoring the importance of adaptive, interactive, and personalised learning environments in promoting equity, accessibility, and academic achievement for students with ADHD.

2 Related work

2.1 ADHD in higher education and traditional educational limitations

ADHD is a prevalent neurodevelopmental disorder characterised by persistent patterns of inattention, hyperactivity, and impulsivity, significantly affecting students' academic and organisational skills in higher education environments. The prevalence of ADHD among university students is estimated to range between 5% and 10%, highlighting the importance of adequately addressing the educational needs of this substantial student population [10]. University students with ADHD frequently struggle with tasks that demand sustained attention, effective time management, and structured organisation, presenting notable barriers to their academic success and personal development [11, 12].

The cognitive and behavioural profiles associated with ADHD often lead to challenges within traditional educational settings, particularly those characterised by passive learning methodologies [13]. Common instructional methods in higher education, such as large-group lectures, conventional reading assignments, and standardised testing, are typically designed without consideration for diverse cognitive needs [14]. Such approaches require prolonged attention spans, robust executive functioning skills, and consistent self-regulation, areas where ADHD students often experience substantial difficulties. Consequently, these conventional teaching methods can inadvertently exacerbate issues related to distractibility, reduced motivation, and frustration, culminating in suboptimal academic outcomes for students with ADHD. Research indicates that students with ADHD consistently demonstrate lower academic performance compared to their neurotypical peers, experiencing higher dropout rates and reduced levels of academic achievement [15–17]. Traditional educational strategies predominantly adopt a one-size-fits-all approach, offering limited flexibility and minimal opportunities for personalisation. This rigid structure poses particular challenges for ADHD students, whose cognitive processing and attentional capabilities fluctuate significantly, even within short timeframes.

Furthermore, standard organisational and study tools commonly provided by educational institutions, such as paper planners, basic digital calendars, and noninteractive e-learning modules, are often inadequate for addressing the specific needs of ADHD students [18, 19]. These tools typically lack the interactive, engaging, and adaptive features essential for maintaining the attention and motivation of ADHD learners. Consequently, the utilisation of conventional organisational resources often leads to increased cognitive load, frustration, and decreased overall academic performance among students with ADHD.

The limitations of traditional educational methods highlight an urgent need for innovative approaches that accommodate diverse cognitive profiles [20, 21]. Educational interventions that leverage interactive and personalised methodologies are essential for effectively supporting students with ADHD. Recent studies underline the effectiveness of active learning strategies that engage students in dynamic, participatory activities rather than passive information consumption [22–24]. These strategies facilitate

sustained attention, deeper cognitive engagement, and improved retention of academic content [23, 25, 26]. Additionally, the higher education context poses unique challenges that exacerbate ADHD-related difficulties [27, 28]. Unlike earlier educational stages, university settings require greater levels of autonomous learning, self-directed study, and effective personal management. These demands necessitate robust executive functioning abilities, such as self-motivation, task initiation, and goal-directed behaviour, skills that are frequently compromised in ADHD individuals. Consequently, the transition to university can significantly intensify existing challenges, leading to heightened academic anxiety, decreased self-efficacy, and compromised mental health among students with ADHD [29–31].

Educational support for university students with ADHD has traditionally focused on generic accommodations, such as extended time on examinations, alternative testing environments, and note-taking support. While these measures can partially alleviate some barriers, they often fail to address underlying cognitive and attentional challenges comprehensively. Therefore, there remains a critical gap in effectively supporting these students through tailored educational approaches that foster essential academic skills and cognitive development. Given these limitations, research increasingly advocates for the integration of technology-enhanced learning solutions tailored specifically for students with ADHD [32–34]. Digital educational technologies hold substantial promise in addressing the educational needs of ADHD learners by providing adaptive, interactive, and personalised learning experiences. These technologies can dynamically adjust content delivery methods, adapt instructional complexity based on real-time performance data, and actively engage students through interactive learning activities [35, 36].

2.2 Augmented reality (AR) in supporting students with ADHD

AR has emerged as an innovative educational technology capable of significantly improving learning outcomes, particularly for students with ADHD. AR technology merges digital content with the physical environment, creating immersive and interactive learning experiences that actively engage learners and enhance their cognitive processing capabilities. This interactive nature of AR addresses several core challenges experienced by students with ADHD, such as sustained attention, task engagement, and comprehension of complex or abstract concepts [37, 38]. AR applications have been effectively utilised in educational contexts to visually represent and simplify complex academic concepts, thus aiding comprehension and reducing cognitive overload. The interactive visualisations offered by AR transform abstract or theoretical content into tangible and engaging experiences, helping students with ADHD maintain focus and actively participate in their learning. Research conducted by Huang et al. [6], demonstrated that AR-based educational platforms significantly improved students' engagement levels, reduced distractibility, and promoted deeper cognitive engagement compared to traditional learning methods.

Moreover, AR technology facilitates multimodal learning by integrating visual, auditory, and kinaesthetic elements into educational experiences. This multimodal approach caters specifically to the diverse learning preferences and cognitive strengths of ADHD learners, who often benefit from dynamic, hands-on, and visually enriched educational environments. Martin et al. [7] emphasise that the multimodal features of AR significantly enhance attention retention, improve information processing, and foster greater

academic motivation among students diagnosed with ADHD. Gamification elements commonly incorporated into AR educational platforms further contribute to their efficacy in supporting students with ADHD. By leveraging principles of game design, such as immediate feedback, rewards, interactive challenges, and structured progress tracking, AR-based gamification promotes sustained engagement and intrinsic motivation. These game-like features effectively address the motivational and attentional challenges characteristic of ADHD learners, encouraging continuous and active participation in educational tasks. Additionally, AR technologies can be seamlessly integrated with adaptive learning systems to personalise educational experiences further. By combining AR with adaptive learning algorithms, educational platforms can dynamically adjust the complexity and pacing of content according to individual student needs, preferences, and performance metrics [39]. This level of personalisation is particularly beneficial for students with ADHD, whose cognitive and attentional needs are highly variable and require flexible, individualised instructional approaches.

2.3 Generative AI and personalised adaptive learning

GenAI represents a major advancement in personalised adaptive learning, providing highly individualised educational experiences tailored to diverse learner profiles, particularly benefiting students with ADHD. By leveraging advanced machine learning algorithms, GenAI systems dynamically adapt educational content, complexity, pacing, and instructional strategies based on continuous real-time analysis of learner interactions, performance data, and behavioural patterns. This adaptive approach directly addresses the fluctuating attention spans, varied cognitive needs, and executive functioning challenges characteristic of ADHD students [8, 40]. The application of GenAI in education includes intelligent tutoring systems, adaptive assessment tools, and personalised content recommendation engines. These tools offer immediate, targeted feedback, enabling students to understand and correct errors promptly, thus maintaining their engagement and motivation. Research by Slavych et al. [9] has highlighted how GenAI-driven chatbots like ChatGPT can significantly enhance reading comprehension among ADHD learners by providing summaries, simplifications, and key concept identifications, thereby reducing cognitive load and enhancing text retention.

GenAI's adaptive personalisation is especially effective in managing ADHD-related cognitive variability. By continuously monitoring a learner's interactions, GenAI systems can accurately predict and adapt to moments of decreased attention or increased distractibility, offering tailored support precisely when needed. This targeted approach significantly enhances learning efficiency, reduces frustration, and supports sustained engagement in educational activities. Furthermore, GenAI facilitates multimodal educational delivery, combining visual, auditory, and textual content to accommodate diverse learning preferences, further enhancing comprehension and retention for students with ADHD. Integrating GenAI with other educational technologies, such as AR, amplifies the benefits of personalised adaptive learning. Such integrations create immersive, interactive environments enriched by real-time adaptive feedback, significantly improving learner engagement and educational outcomes.

While the potential of AR and GenAI technologies in education is well-documented, not all studies report uniformly positive outcomes. For instance, Akçayır and Akçayır [41] and Garzón et al. [42] note that technical complexity, novelty effects, and

inconsistent implementation fidelity can limit the long-term impact of AR in learning environments. Similarly, Barua et al. [34] and Gligorea et al. [39] caution that personalised AI tools may reinforce surface learning or result in reduced learner autonomy when overused. A recent meta-analysis by Buchner et al. [5] found that AR's impact on cognitive load varies significantly depending on task complexity and learner characteristics. These findings underscore the importance of contextual, learner-sensitive implementation, particularly for neurodiverse populations. By critically engaging with these contrasting perspectives, this study aims to both build on and scrutinise prevailing assumptions in the field of technology-enhanced ADHD education.

While previous studies have explored the use of AR or GenAI independently to support learners with ADHD, few have examined their combined potential in a higher education context. Most existing interventions have focused on younger learners (K–12), with limited attention to the unique cognitive and organisational demands faced by university students. Additionally, prior work often lacks detailed analysis of how these technologies can be tailored to support fluctuating attention spans and executive dysfunctions in real-world academic settings. This study seeks to address these gaps by investigating a hybrid AR–GenAI solution explicitly designed for ADHD students in higher education, grounded in both technical and cognitive accessibility principles.

3 ADHDvance learnAR system design and implementation

3.1 System design and implementation

3.1.1 Design principles and user-centred rationale

ADHDvance LearnAR was developed using a user-centred design (UCD) approach that explicitly aligns system features with the cognitive and behavioural characteristics commonly associated with ADHD. Prior research highlights that students with ADHD often experience executive dysfunction (e.g., planning, working memory, task initiation), heightened distractibility, and fluctuating attention. In response, the design prioritised three guiding principles: simplicity (to minimise cognitive load and reduce distraction), interactivity (to maintain attention through active engagement), and personalisation (to support diverse needs and changing attention states across tasks and contexts).

User involvement was embedded throughout the development process through iterative design cycles rather than confined to end-stage evaluation. Students with ADHD contributed practical feedback on which features were most useful, where friction occurred, and what interface elements felt distracting or overwhelming. This input shaped key decisions such as reducing visual clutter, ensuring predictable navigation, and limiting the number of steps required to create tasks, trigger reminders, or access support tools. Educators also participated by advising on classroom feasibility, workload implications, and alignment with established study-skills and support practices in higher education.

These participatory inputs directly informed the system's final design: a minimalist interface intended to reduce distraction; short, actionable prompts to support task initiation and reduce decision fatigue; and modular feature access to allow learners to use only the supports they need at a given moment. Together, these design principles and stakeholder contributions ensured that ADHDvance LearnAR was not only technically functional, but also pedagogically appropriate, cognitively accessible, and grounded in the lived experiences of its intended users.

3.1.2 Executive function support: task management and visual planning

To address challenges in organisation, time management, and task initiation, ADHDvance LearnAR integrates visual-spatial task management features using Augmented Reality. The AR To-Do List and Calendar present tasks and events as anchored 3D elements within the learner's environment. These affordances enhance the salience of pending responsibilities and support cognitive offloading, a key benefit for students with working memory limitations. By visually externalising goals, students can more effectively prioritise and monitor progress. Integration with colour-coded urgency levels and visual timelines helps mitigate time-blindness, a common executive function difficulty among ADHD populations.

3.1.3 Sustained attention and cognitive load reduction

Recognising that learners with ADHD often experience reduced attention spans and heightened distractibility, ADHDvance LearnAR includes features designed to scaffold focus and reduce cognitive overload. The Focus Time AR tool implements the Pomodoro Technique, breaking work into structured intervals with interspersed breaks. This technique is known to support attention regulation and task endurance. Visual countdowns and auditory transitions further aid temporal orientation. Additionally, the system offers Bionic Text Reading Mode, which emphasises key information within summarised content, reducing cognitive strain during reading and enhancing information retention.

3.1.4 Adaptive learning and personalised feedback

The integration of Generative AI within ADHDvance LearnAR enables highly adaptive and responsive educational experiences. Through tools such as the AI Summariser and the ADHDvance Chatbot, the system delivers real-time personalised assistance, including content simplification, study recommendations, and organisation prompts. These tools respond to students' behavioural patterns, learning preferences, and prior interactions, providing just-in-time support without overwhelming the learner. The system's adaptive components are particularly valuable in helping students sustain momentum and overcome executive dysfunction by providing structure, encouragement, and corrective feedback tailored to their individual needs.

3.1.5 User experience and accessibility design

The UI/UX design of ADHDvance LearnAR was intentionally minimalist to align with the cognitive and behavioural characteristics of students with ADHD. Individuals with ADHD often experience heightened distractibility and sensory sensitivity, making visual simplicity and predictability essential for maintaining focus. To reduce cognitive load, the interface avoids unnecessary animations, dense text blocks, and excessive visual clutter. Consistent iconography, limited colour palettes, and straightforward navigation paths support goal-directed behaviour and reduce the demands on working memory and attention-switching. Feedback from ADHD users during prototyping phases confirmed the value of clean layouts, chunked information, and simple interactive elements. These design decisions reflect established accessibility guidelines for neurodiverse learners and ensure that the technology supports, not overwhelms, students with attentional and executive function challenges.

4 Implementation

The implementation of ADHDvance LearnAR was a complex and multi-faceted process that involved the integration of cutting-edge technologies to create a functional, user-friendly educational tool. This section details the implementation of the key features, focusing on the technical challenges encountered and the solutions developed to address them.

4.1 AR To-Do list and AR calendar

The AR To-Do List and AR Calendar were among the first features implemented in ADHDvance LearnAR. To support executive function challenges common among ADHD students, ADHDvance LearnAR incorporates AR-based task visualisation tools, including a To-Do List and Calendar. These features present tasks as spatially anchored 3D elements, enhancing planning and time management through visual reinforcement. By linking academic deadlines to immersive prompts, students experience improved task salience and reduced working memory burden, challenges frequently cited in ADHD literature [10, 35].

During the implementation of these features, one of the primary challenges was ensuring that the AR elements were intuitive and did not overwhelm the user. Given that ADHD students can be easily distracted, it was crucial to design an interface that is simple yet effective in helping them manage their tasks. The solution was to adopt a minimalist design approach, focusing on essential functions and using colour coding and simple icons to help students quickly identify and prioritise tasks. This approach ensured that the interface remained user-friendly while providing the necessary tools for effective task management.

4.2 Focus time AR

Focus Time AR was developed as a feature to enhance students' focus and time management through the Pomodoro Technique. For focus and self-regulation, the system embeds a Focus Time AR tool modelled on the Pomodoro Technique. This design scaffolds attention by breaking learning into manageable chunks, paired with adaptive visual and auditory cues. Such structured pacing has been shown to support sustained attention and mitigate cognitive fatigue in students with ADHD [41]. This system was designed to be engaging without being distracting, helping students maintain a balanced study routine.

The main challenge during the implementation of Focus Time AR was to design a timer that is effective in helping students focus without causing frustration or distraction. The solution was to incorporate clear indicators of time remaining in each interval, along with gentle auditory cues to signal transitions between work and break (Figs. 1, 2, 3, 4, 5)

periods. The timer's design is simple, with an emphasis on clarity and ease of use, ensuring that students can use it without feeling overwhelmed.

4.3 AR document scan and AI summarizer

The AR Document Scan and AI Summarizer feature is another critical component of ADHDvance LearnAR. This feature uses AR technology to scan documents in real-time, with GenAI providing concise summaries. The implementation of this feature involved

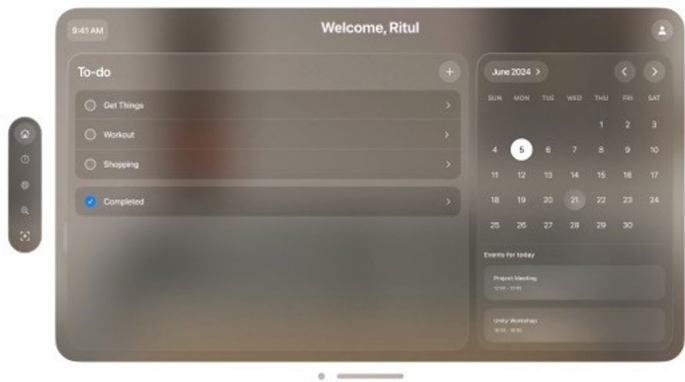


Fig. 1 Home Page: AR To-Do List and AR Calendar

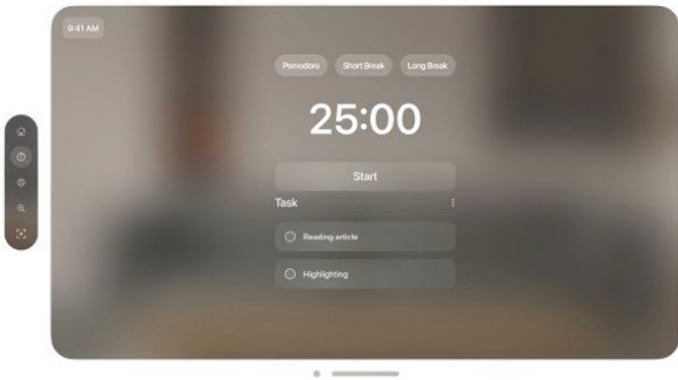


Fig. 2 Focus time AR: pomodoro timer

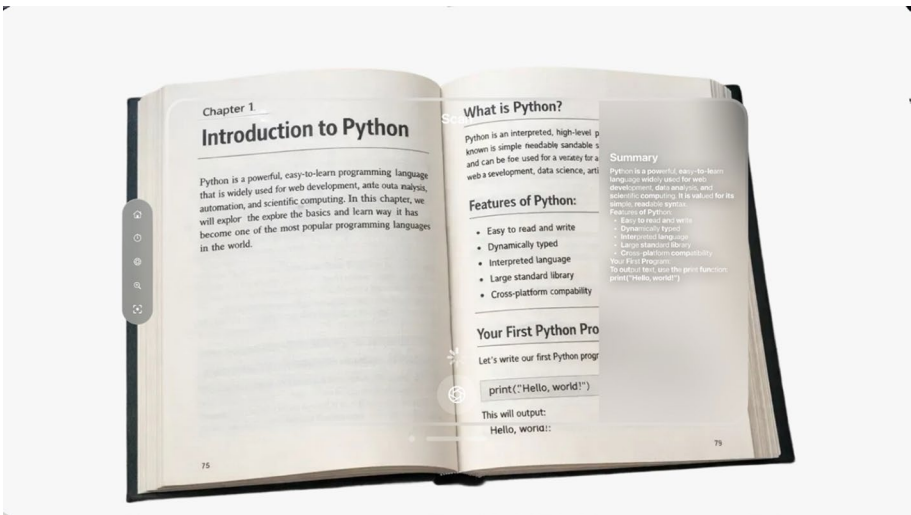


Fig. 3 ADHDvance LearnAR document scan and GenAI summariser (anonymised example output), including simplified summary and reading-support formatting



Fig. 4 ADHDvance LearnAR GenAI chatbot (anonymised example interaction) providing just-in-time scaffolding for study guidance



Fig. 5 AR 3D education: interaction with 3D models in AR

developing algorithms capable of accurately scanning and summarising various types of documents, while maintaining the readability of the output, particularly in the Bionic Text Reading mode.

One of the key challenges in implementing this feature was ensuring that the document scanning and summarisation processes were both reliable and user-friendly. The solution was to fine-tune the scanning algorithms to handle different document layouts and content types, ensuring accuracy and efficiency. Additionally, the Bionic Text Reading feature was integrated with customisable settings, allowing students to adjust the text format to suit their preferences, thereby enhancing focus and retention.

4.4 ADHDvance learnAR AI chatbot

The ADHD AI Chatbot was implemented to provide students with personalised support in organising their work, managing tasks, and answering questions. The chatbot is powered by GenAI, which was trained on a diverse dataset of educational queries to provide

accurate and helpful responses. The implementation of the chatbot involved creating a conversational interface that is responsive, informative, and easy to interact with.

The primary challenge during the implementation of the ADHDvance LearnAR AI Chatbot was ensuring that the bot could understand and respond accurately to a wide range of student inquiries. To address this challenge, the chatbot's GenAI model was extensively trained and tested to handle various educational scenarios. The design of the chatbot interface was kept conversational and friendly, encouraging students to interact with the bot as they would with a human tutor. This approach helps reduce the reliance on teacher intervention, empowering students to take control of their learning.

4.5 AR 3D education

The AR 3D Education feature was developed to make abstract concepts more tangible by presenting educational content in the form of 3D models. This feature allows students to interact with 3D representations of various educational subjects, such as geography, biology, and physics. The implementation involved creating detailed 3D models that are both educationally valuable and engaging for ADHD students.

The challenge in implementing AR 3D Education was to create models that were visually appealing without overwhelming the user. The solution was to design the 3D models with simplicity and educational purpose in mind, ensuring that they are easy to interact with and understand. Interactive elements were added to the models to encourage exploration, while visual complexity was minimised to prevent cognitive overload.

4.6 Design and implementation decisions

Several critical decisions were made during the design and implementation of ADHDvance LearnAR to ensure the application's success and effectiveness.

User-Centered Design: The design process was heavily guided by feedback from ADHD students. This user-centered approach ensured that the app is both functional and accessible, addressing the specific needs and preferences of its target users. By involving end-users in the design process, the app was tailored to effectively support ADHD students in their learning journey.

Modular Design: The app's architecture was designed to be modular, allowing for scalability and future enhancements. This decision was crucial in ensuring that the app can evolve with advancements in AR and AI technologies, providing long-term value to its users. The modular design also facilitates the easy addition of new features and improvements, making the app adaptable to changing educational needs.

Data Security and Privacy: Given the sensitive nature of the data involved, strong data security protocols were implemented, including encryption and secure storage practices. This decision was essential for maintaining user trust and ensuring compliance with data protection regulations. Ensuring the privacy and security of user data was a top priority throughout the design and implementation process, reflecting the app's commitment to ethical standards in educational technology.

5 Methodology

This study employed a Design-Based Research (DBR) methodology to iteratively develop, implement, and evaluate ADHDvance LearnAR, an educational application integrating AR and GenAI, specifically designed to support university students

diagnosed with ADHD. DBR was chosen due to its suitability for educational technology innovations, combining rigorous theoretical inquiry with practical intervention development and promoting iterative refinement based on continuous evaluation and user feedback [43, 44].

5.1 Participants

The study involved 15 undergraduate students diagnosed with ADHD, recruited through purposive sampling at a UK university. While modest, this sample size aligns with best practices in early-phase Design-Based Research (DBR), where the focus is on iterative development, contextual validation, and in-depth user feedback [43, 44]. Given the exploratory nature of the study, the aim was not statistical generalisability but the identification of meaningful patterns and user-informed design improvements. Nevertheless, this limitation is acknowledged in our discussion of findings and implications.

5.2 Research design

The study adopted a within-subjects experimental design comprising three distinct sequential conditions to comprehensively evaluate the effectiveness of the ADHDvance LearnAR application. Initially, participants experienced a baseline condition where they employed traditional educational tools and organisational methods, such as physical planners and conventional digital calendars. This baseline provided a comparative standard to measure the effectiveness of subsequent interventions. Subsequently, in the AR-Enhanced condition, participants engaged with the ADHDvance LearnAR application, exclusively utilising its AR features, including interactive modules and tasks designed to support learning and organisation. In the final Combined AR + GenAI condition, participants interacted with the fully integrated ADHDvance LearnAR application, incorporating both AR visualisation modules and GenAI-driven adaptive and personalised educational support. The sequential experience of these conditions allowed for controlled, within-subject comparisons to effectively evaluate the distinct and combined impacts of AR and GenAI technologies. To mitigate potential confounding variables in the within-subjects design, the order of experimental conditions was counterbalanced across participants. Furthermore, all participants completed equivalent tasks during each condition, and sessions were conducted in a controlled classroom environment to minimise external distractions. The use of standardised instructions and identical timing across sessions further ensured internal validity.

5.3 Research phases

Consistent with the Design-Based Research (DBR) framework, this research was structured into four iterative and interconnected phases, beginning with the analysis and exploration phase. In this initial stage, core educational challenges encountered by university students with ADHD were identified through extensive literature reviews and preliminary consultations involving stakeholders such as students and educators. Insights gained from these activities provided a solid foundation for developing the conceptual framework and design principles underlying the ADHDvance LearnAR application. Following this was the design and development phase, which leveraged the findings from phase one. This second phase entailed iterative processes of prototyping and refining ADHDvance LearnAR, embedding interactive AR visualisation tools and adaptive

GenAI functionalities specifically crafted to support executive functioning, sustained attention, and personalised learning experiences. Throughout this process, ongoing stakeholder feedback was actively incorporated to ensure the system's practical relevance and optimal usability.

Subsequently, the implementation and testing phase involved rigorous evaluation of ADHDvance LearnAR across multiple sessions. Each participant engaged with the distinct experimental conditions during three separate sessions, each lasting approximately 90 min, conducted within a controlled classroom environment. Standardised tasks were employed consistently across all conditions to enable valid and reliable comparisons of the system's effectiveness. The final phase, evaluation and refinement, involved systematic collection and analysis of data from the testing sessions. The findings from this phase informed additional iterative refinements, ensuring the final iteration of ADHDvance LearnAR was closely aligned with user needs and effectively enhanced educational outcomes and usability for university students with ADHD.

5.4 Data collection instruments

A mixed-methods approach was used to capture both behavioural indicators of engagement and participants' subjective experiences of using ADHDvance LearnAR. Quantitative data were collected via the application's built-in analytics, which automatically logged objective engagement metrics including active session duration, task interaction frequency, and task completion rates. These measures are widely used in educational technology research as indicators of engagement and learning-related behaviour. In addition, participants completed a standardised study-skills instrument, the Learning and Study Strategies Inventory (LASSI) [45], to assess perceived academic and organisational strategies across conditions (e.g., time management, concentration, and strategic learning).

Qualitative data were collected to provide explanatory depth regarding how and why the AR and GenAI features shaped learners' experiences. Semi-structured interviews were conducted after the intervention with five participants (from the full sample of 15) to explore perceived usefulness, usability, cognitive load, and the ways the tools influenced task initiation, attention regulation, and organisation. Interviews lasted approximately 30–43 min, were audio-recorded with consent, and transcribed for analysis. To complement the interviews and capture perspectives from the full cohort, all 15 participants completed a post-intervention questionnaire comprising Likert-scale items and open-ended questions. Questionnaire items were informed by validated usability instruments, including the System Usability Scale (SUS), enabling systematic assessment of perceived ease of use, satisfaction, and areas for improvement. Together, these quantitative and qualitative instruments supported triangulation of observed behavioural trends with participant-reported experiences.

5.5 Data analysis

Quantitative data were analysed to characterise engagement and performance across the three within-subject conditions and to examine whether observed differences were statistically reliable. We first summarised each outcome using descriptive statistics (means and standard deviations). We then applied repeated-measures ANOVA to test

Table 1 Descriptive statistics for engagement, interaction frequency, and task completion rates

Condition	Engagement duration (min)		Interaction frequency		Task completion rate (%)	
	Mean	SD	Mean	SD	Mean	SD
Baseline	40.67	4.49	10.37	1.61	61.60	7.40
AR-Enhanced	54.47	5.30	19.83	3.69	75.57	7.45
AR+GenAI	69.32	4.46	30.27	3.94	84.80	4.28

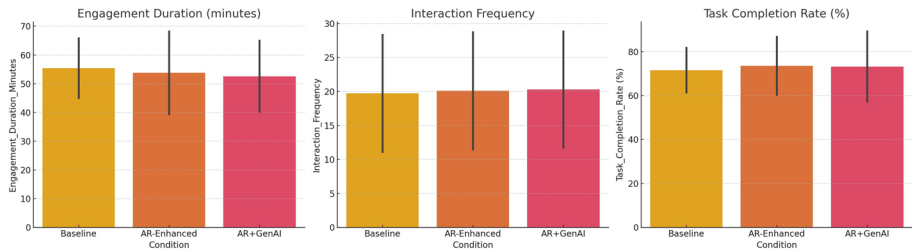


Fig. 6 Task completion rates across experimental conditions

for condition effects, followed where appropriate by pairwise post-hoc comparisons to explore differences between baseline, AR-enhanced, and AR+GenAI conditions.

Qualitative data from semi-structured interviews and open-ended questionnaire responses were analysed using an inductive, structured thematic coding approach informed by Braun and Clarke [46]. Transcripts and text responses were coded for meaning units relevant to the research questions, codes were iteratively refined and grouped into descriptive categories, and categories were synthesised into higher-level themes. To support transparent reporting, we summarised theme prevalence (counts/percentages) and selected representative anonymised quotations to illustrate how themes were grounded in participant accounts. Finally, we integrated quantitative and qualitative findings through triangulation, using qualitative themes to explain, contextualise, and qualify behavioural trends observed in the quantitative metrics.

6 Findings

6.1 Research question 1: How do AR visualisations and interactive features influence engagement and task completion among university students with ADHD?

While no statistically significant differences were found across conditions ($p > 0.05$), descriptive statistics indicated a positive trend in engagement and task completion when AR visualisations were introduced. Engagement duration increased from 40.67 min at baseline to 54.47 min in the AR-enhanced condition, while interaction frequency and task completion showed similar upward trends (Table 1).

Although preliminary, these trends suggest that AR features may contribute to improved focus and task follow-through for students with ADHD. Task completion rates also showed improvement, increasing from a mean of 61.60% (SD = 7.40) at Baseline to 75.57% (SD = 7.45) with AR enhancements (Fig. 6).

Although these improvements did not reach statistical significance (see Table 1, 2), qualitative feedback supported this interpretation, with 80% of students reporting moderate to high engagement and noting the motivational benefits of interactive AR elements (Table 3).

Table 2 ANOVA results for engagement metrics, interaction frequency, and task completion rates

Metric	F-Value	Degrees of Freedom	p-value
Engagement Duration	1.39	(2, 28)	0.265
Interaction Frequency	0.12	(2, 28)	0.886
Task Completion Rate	0.35	(2, 28)	0.708

Table 3 Summary of qualitative themes and prevalence across participant responses (interviews and open-ended items). Representative quotations are reported in Sect. 6.5

Theme	Response category	Percentage (%)
Engagement	Highly engaged and enjoyed interactive elements	35
	Moderately engaged but had trouble maintaining focus	45
	Felt disengaged and easily distracted	20
Task organisation	Experienced marked improvement in task organisation	40
	Managed tasks better but still faced difficulties	45
	Struggled significantly with task management	15
Usability	Found the system intuitive and very easy to use	30
	The system was user-friendly but had occasional issues	50
	Found the system difficult to use	20
AI personalisation	Found AI-driven personalisation very helpful	30
	Felt somewhat supported by AI-driven prompts	35
	Did not feel any personalised support	35

6.2 Research question 2: what effects do generative AI-driven personalised learning experiences have on the organisational skills and academic performance of students with ADHD?

The addition of GenAI-based personalisation to the AR environment was associated with further gains in observed engagement and task performance, although again without statistical significance. Engagement time rose to 69.32 min, and task completion rates reached 84.80% (SD = 4.28). While these patterns must be interpreted cautiously due to sample size and lack of significance, qualitative data from 65% of participants described enhanced clarity, direction, and motivation attributed to personalised AI support (Table 3). This convergence of subjective and behavioural data points to the potential value of GenAI in scaffolding organisational skills for ADHD learners. Interaction frequency likewise improved notably from 19.83 interactions (SD = 3.69) to 30.27 interactions (SD = 3.94).

6.3 Research question 3: can the integration of AR and GenAI technologies effectively reduce cognitive overload and manage behavioural symptoms such as impulsivity and inattention in students with ADHD?

The integration of AR and GenAI technologies yielded promising qualitative insights regarding their potential to manage cognitive overload and behavioural symptoms. Participants frequently reported reduced cognitive strain due to the interactive and personalised elements of the combined approach (Table 3). While quantitative metrics such as engagement duration and interaction frequency suggested improved attention and reduced impulsivity indirectly, these changes were not statistically significant (Table 1). Nevertheless, qualitative feedback strongly indicated perceived benefits, with participants emphasising that the interactive and personalised interventions enhanced their capacity to sustain attention, organise tasks, and reduce impulsive behaviours during learning activities.

6.4 Research question 4: what implications do the combined use of AR and GenAI technologies have on students' long-term academic and social outcomes?

The data collected through qualitative thematic analysis highlighted potential longterm academic and social benefits from the combined use of AR and GenAI technologies. Participants expressed optimistic views about sustained academic improvement and better management of social interactions stemming from enhanced organisational skills and reduced anxiety related to academic tasks (Table 3). Despite quantitative analyses not achieving statistical significance, the clear positive trends, coupled with qualitative feedback, underscored the potential implications for long-term academic confidence and social competency among students with ADHD. The qualitative responses particularly emphasised improved confidence in social interactions, driven by enhanced academic performance and organisational proficiency facilitated by AR and GenAI support.

Although none of the observed quantitative differences reached statistical significance, the directional consistency of the trends across conditions, coupled with strong qualitative validation, supports the exploratory nature of this study. These findings should be considered indicative rather than conclusive, highlighting areas for further inquiry rather than definitive outcomes.

6.5 Qualitative findings (thematic synthesis)

Qualitative analysis provided explanatory depth on how and why participants experienced the AR and GenAI features as supportive, complementing the directional trends observed in engagement, interaction frequency, and task completion. Across the interview corpus, participants repeatedly described the tool as reducing the “friction” associated with executive dysfunction, particularly difficulties with time perception, task initiation, sustained attention, and managing cognitive overload. Four themes were identified, each illuminating a distinct mechanism through which AR and GenAI were perceived to influence study behaviour.

6.6 Theme 1: AR increased task salience and supported planning through externalisation

Participants consistently described AR task visualisation as counteracting the “out of sight, out of mind” experience that often accompanies ADHD. Rather than relying on internal reminders, tasks became external objects that remained visible in the environment, supporting planning through continual low-effort prompting. One participant likened the AR tasks to persistent reminders that could not be misplaced: *“It’s like a digital sticky note that I actually can’t lose under a pile of papers”* (P05). This sense of permanence was repeatedly framed as reducing the cognitive burden of remembering what needed to be done. As P12 explained, *“Usually, if I close my planner, the tasks cease to exist in my brain. Having the deadlines literally floating over my desk kept them in my ‘working memory’ without me having to use any actual mental energy to remember they were there.”*

Several participants also described how spatial anchoring supported planning by enabling them to structure attention and separate competing demands. For example, P03 reported using physical space as an organisational scaffold: *“The spatial anchoring is the best part. I associated the ‘AR window’ on the left with my research and the one on the right with my draft. It partitioned my physical space for my brain.”* This suggests that AR

did more than display tasks; it allowed participants to externalise an internal organisational strategy by mapping academic subtasks onto distinct spatial locations.

Importantly, AR visualisation was also perceived to influence self-regulation by increasing the immediacy of deadlines and reducing avoidance behaviours. Participants described the AR overlays as creating a sense of accountability that was harder to escape than conventional digital tools. P08 noted, *“It made the deadline feel like a real, physical thing in the room with me. I couldn’t just scroll on my phone because the task was literally hovering right there, staring at me.”* Relatedly, time perception—frequently described as distorted or “time-blind”—was reported to improve when time cues were rendered spatially. P14 observed, *“Seeing the countdown timer in 3D space made time feel like it was actually passing. Usually, I’m totally time-blind, but this made the ‘now’ feel much more urgent.”* Collectively, these accounts suggest that AR-based externalisation increased task salience, supported prioritisation, and provided a practical countermeasure to time-blindness and working-memory overload.

6.7 Theme 2: structured pacing supported sustained attention and task initiation

Participants emphasised that the structured pacing mechanisms, particularly the focus-timer workflow, reduced the psychological barrier to starting tasks and supported sustained engagement once work began. Several described the familiar experience of “task paralysis” or avoidance at the point of initiation, and framed the tool as providing an initial push that lowered the activation energy required to begin. P01 captured this succinctly: *“The focus-timer helped me climb the ‘wall of awful’ that usually stops me from starting.”*

Others explained that the intervention worked by reducing perceived task magnitude. Rather than approaching study as a single, indefinite block of effort, participants experienced the work as a sequence of short, bounded intervals. P09 described a shift from overwhelm to manageability: *“I’m a chronic procrastinator because starting feels like a mountain. But the way the AR system paced the intervals made it feel like I was just doing a series of small, five-minute sprints instead of a marathon. I actually finished the whole session.”* This account suggests that pacing supported task initiation by reframing work into achievable units, a mechanism that aligns closely with executive function supports recommended for ADHD learners.

Sustained attention was also described as being supported through non-intrusive prompts that remained perceptible without becoming disruptive. P06 noted that attention could be redirected without loud alarms or interruptions: *“It kept me from drifting off. If I looked away to check my phone, the visual cue in my peripheral vision reminded me that I was still ‘on the clock’ without being annoying or loud.”* This subtle “attention tether” was experienced as particularly valuable because it preserved focus while minimising irritation, an important consideration for learners who may be sensitive to overstimulation.

Participants further described improvements in persistence and reduced off-task movement. P11 stated, *“I actually stayed in my chair for 50 minutes. That’s a record for me without getting up to make coffee or getting distracted by something else.”* Finally, transitions between work and breaks were highlighted as a crucial design feature. For some participants, breaks can become derailments; returning to the task is often harder than starting. P04 explained that the system supported re-engagement: *“The transitions were*

key. Usually, when I take a break, I never come back. The system gave me a clear 're-entry' prompt that made it easy to pick up the thread of what I was writing." Taken together, these accounts suggest that structured pacing operated through two related mechanisms: (i) reducing initiation barriers by shrinking tasks into manageable intervals, and (ii) sustaining attention through predictable, low-friction prompts that support re-entry after breaks.

6.8 Theme 3: GenAI scaffolding reduced friction and improved clarity of next steps

Participants frequently characterised GenAI features as reducing "stuckness" during complex academic work, particularly when they were uncertain how to begin, how to organise information, or what the next actionable step should be. Rather than describing GenAI as providing answers, participants emphasised its role in offering an "on-ramp" into tasks. P15 stated, *"It gave me an 'on-ramp' when I was stuck on the first sentence."* This framing recurred across interviews: GenAI support was valued when it transformed vague intention into a concrete starting action.

Decision fatigue and planning were recurring pressure points. P02 described how supportive scaffolding reduced anxiety and enabled momentum: *"The AI was like having a tutor who doesn't judge you for asking the same question three times. When I didn't know how to organize my notes, it gave me three bullet points to follow, and that was enough to get me moving again."* Here, the perceived value lies in turning an unstructured cognitive load into a small set of steps, sufficient structure to proceed without overwhelming the learner.

Participants also valued prompts that encouraged strategic approaches rather than content generation. P10 highlighted the importance of "nudges" that guide attention and reduce overwhelm: *"I liked that it didn't just give me the answer. It gave me a 'nudge.' It was like, 'Hey, maybe try looking at the conclusion of this paper first,' which stopped me from feeling overwhelmed by the 20 pages I had to read."* This illustrates a key pedagogical implication: GenAI may be most helpful for ADHD learners when it supports metacognitive strategy use (where to start, how to skim, how to prioritise), not when it replaces thinking.

Similarly, P07 described GenAI as enabling task entry when motivation and planning were misaligned: *"Sometimes I know what I want to say, but I can't find the 'door' to the task. The GenAI prompts basically opened the door for me so I could just start walking."* Finally, several participants emphasised that the effectiveness of GenAI support was amplified by its situated integration within the AR workflow. Rather than feeling generic, GenAI responses were experienced as context-sensitive and embedded in the moment of work. P13 explained, *"It's way better than a regular chatbot because it was specific to what I was looking at in the AR view. It felt like it was in the trenches with me."* Overall, Theme 3 suggests that GenAI served as a practical mechanism for reducing decision fatigue, providing actionable next steps, and enabling task initiation without undermining learner agency.

6.9 Theme 4: perceived cognitive load decreased when content was simplified and chunked

Participants frequently reported that the system reduced cognitive load, particularly during reading and interpretation of dense materials, by simplifying, chunking, and

filtering information. Many described experiencing reading fatigue, “brain fog,” and frustration when confronted with long texts. In this context, summarisation and structured outputs were experienced as transforming materials from overwhelming to manageable. P08 described this shift succinctly: *“It turned a ‘wall of text’ into something I could actually digest.”*

For some, the value of summarisation lay in reducing repeated rereading and mental exhaustion. P05 reported: *“Reading long papers is my nightmare; I usually have to read the same paragraph four times before it clicks. The AI summary feature meant I actually understood the main point on the first go, which saved me so much mental exhaustion.”* This reflects both a cognitive efficiency benefit (fewer rereads) and an affective benefit (reduced frustration), which can be particularly important for sustaining engagement over time.

Participants also described “chunking” as reducing stress responses that can trigger avoidance or shutdown. P03 explained: *“The ‘chunking’ made it feel less like a heavy lift. I could see the whole task broken down into tiny steps, and my brain didn’t immediately go into ‘shutdown mode’ from the stress of it all.”* Others emphasised relief from cognitive strain and physical symptoms associated with overloaded study sessions. P12 noted: *“I didn’t finish the session with a headache for once. By simplifying the language of the instructions, it felt like the system was doing the ‘filtering’ work that my ADHD brain usually struggles to do.”*

Finally, participants described how simplified outputs enabled them to direct effort toward learning rather than decoding instructions. P01 stated: *“It took the noise out of the assignment. I could focus on the actual learning instead of just fighting to understand what the instructions were asking me to do.”* Together, these accounts suggest that the GenAI-supported simplification features reduced extraneous cognitive load by filtering and restructuring information, thereby supporting comprehension, reducing fatigue, and sustaining engagement.

7 Discussion

7.1 RQ 1: How do AR visualisations and interactive features influence engagement and task completion among university students with ADHD?

Across the pilot conditions, AR visualisations were associated with directionally positive patterns in engagement and task performance, although inferential tests did not detect statistically significant differences. The descriptive shifts nonetheless suggest that AR may be a promising mechanism for supporting ADHD learners in higher education, particularly when engagement is operationalised through sustained session time, interaction frequency, and completion behaviour. Rather than implying definitive effectiveness, these outcomes indicate feasibility and warrant cautious interpretation consistent with an exploratory DBR evaluation.

Interpreting these patterns alongside participants’ accounts suggests that AR’s value may lie in how it externalises otherwise fragile executive-function processes. In particular, the qualitative data indicate that AR increased task salience and reduced the “out of sight, out of mind” effect typical of ADHD-related executive dysfunction. Participants described AR tasks as persistently visible in their workspace and therefore harder to dismiss or forget, effectively offloading working-memory demands and making deadlines feel immediate. This mechanism-oriented interpretation aligns with prior

work describing AR as beneficial for attention and engagement through multisensory, interactive cues and immersive task representations [47, 48]. For learners with ADHD, whose attentional regulation often fluctuates, the interactive nature of AR may support sustained focus by anchoring attention in the environment rather than requiring continuous internal self-regulation [49]. More broadly, these findings are consistent with literature suggesting that technology-mediated interactivity can increase intrinsic motivation and promote active participation, potentially mitigating behavioural challenges in learning settings [41].

7.2 RQ 2: what effects do generative AI-driven personalised learning experiences have on the organisational skills and academic performance of students with ADHD?

The addition of GenAI to the AR-enhanced learning environment coincided with further indicative gains in behavioural metrics and participant-reported organisation support, though again without statistically significant differences. From an interpretive standpoint, the combined AR+GenAI condition appears to have offered not only engagement support but also decision-structure support, helping participants move from intention to action by clarifying next steps, reducing planning friction, and providing adaptive prompts. This is consistent with the idea that for ADHD learners, barriers to performance are often rooted less in content difficulty and more in executive control demands (planning, sequencing, initiating, and monitoring progress).

These patterns align with research on adaptive and personalised learning technologies that emphasises their potential for supporting learners with executive functioning difficulties [35, 50]. In this study, participants' qualitative accounts suggest that GenAI was most helpful when it provided short, actionable scaffolds (e.g., organising notes into a small set of steps, prompting a start point, or narrowing attention to a manageable subtask). This form of support may be particularly relevant in higher education where independent study, self-scheduling, and unstructured assignments are common and can disproportionately disadvantage students with ADHD. Notably, participants described GenAI less as an "answer engine" and more as a non-judgemental guide that reduced decision fatigue, an interpretation that supports a pedagogically responsible framing of GenAI as scaffolding rather than substitution.

7.3 RQ 3: Can the integration of AR and GenAI technologies effectively reduce cognitive overload and manage behavioural symptoms such as impulsivity and inattention in students with ADHD?

The qualitative findings offer the strongest explanatory evidence for how the combined AR and GenAI approach may influence cognitive load and attention regulation. Participants reported that AR-based structuring (e.g., timers and persistent cues) supported sustained attention and reduced drifting, while GenAI-based simplification reduced the perceived mental effort required to process dense materials. These accounts align with literature emphasising the role of multimodal and interactive supports in reducing cognitive load for learners with attentional challenges [49], and with perspectives that structured, technology-mediated learning environments can provide targeted cognitive scaffolding [41].

In particular, participants described the focus-timer workflow as supporting task initiation and re-entry after breaks, two well-known friction points for ADHD learners.

They also highlighted that non-intrusive cues (e.g., visible prompts in peripheral vision) helped maintain attention without adding irritation or overstimulation, suggesting that the form of feedback matters as much as its presence. GenAI features were frequently described as reducing overwhelm by “chunking” tasks and filtering instructions, mechanisms consistent with reducing extraneous cognitive load and supporting executive functioning. However, while these experiences are compelling, they should be interpreted as exploratory rather than conclusive, especially given the non-significant quantitative results.

The small sample size ($n = 15$) restricts generalisability and limits statistical power, which reduces the likelihood of detecting effects even when meaningful trends exist. This is common in early-stage DBR evaluations, but it reinforces the need for larger-scale validation and designs that better isolate causal contributions. Accordingly, the integrated AR+GenAI approach should be framed as a plausible support strategy with strong user-perceived mechanisms, rather than as a confirmed intervention for reducing ADHD symptom expression in educational settings.

7.4 RQ 4: What implications do the combined use of AR and GenAI technologies have on students’ long-term academic and social outcomes?

Participants’ feedback suggested optimism regarding longer-term academic confidence, reduced anxiety, and improved self-efficacy when using tools that externalise planning and reduce cognitive friction. These perceived benefits are plausible and consistent with research indicating that interventions targeting organisation, study strategies, and cognitive control can support broader psychosocial outcomes for students with ADHD [51–53]. For example, improved task management and reduced overwhelm may translate into higher persistence, less avoidance, and better participation in group work, factors that can influence both academic achievement and social integration. In the present study, participants described feeling more capable of managing academic responsibilities and more confident in engaging with collaborative tasks, implying potential downstream impacts beyond immediate engagement measures.

Nevertheless, these implications remain tentative because the study did not include longitudinal follow-up or objective measurement of longer-term academic and social outcomes. As such, claims about sustained effects must remain carefully bounded. Future work should directly examine whether the mechanisms observed here (enhanced task salience, improved initiation, reduced cognitive load) translate into durable gains in academic performance, retention, wellbeing, and social functioning over a semester or academic year.

7.5 Practical guidance for educators

Educators can incorporate AR and GenAI supports in ways that complement existing inclusive teaching strategies. First, these tools are most effective when aligned to course routines: for example, embedding weekly planning checkpoints where students translate module requirements into small actionable tasks, and using AR task visualisation to make progress visible. Second, GenAI support should be framed as scaffolding for process rather than answer production. Short prompts that help students plan, summarise readings, or generate study checklists can support executive functioning while preserving academic integrity. Third, educators can reduce cognitive load by providing

materials in chunked formats and encouraging students to use summarisation features to create personal study notes, with explicit expectations for attribution and disclosure where relevant. Finally, practical adoption benefits from low-friction onboarding (brief demonstrations, clear permitted-use statements, and fallback non-AR alternatives) so that students can choose supports that match their preferences and sensory needs.

7.6 Alternative explanations, limitations, and contrasting evidence

Despite the promising patterns, alternative explanations must be considered. First, novelty effects may partly account for increased engagement when participants interacted with unfamiliar technology in a short-term setting. Second, structured session conditions and researcher presence may have increased accountability and reduced distractions independent of the intervention. Third, even with procedural controls, within-subject sequencing can introduce learning or carryover effects, complicating attribution.

Methodologically, the pilot scale ($n = 15$) limits statistical power and generalisability. The absence of a comparison condition that excludes all technology-based supports also restricts the ability to isolate intervention-specific effects. These limitations mean that the contribution of this work is best framed as a pilot demonstration of feasibility and design plausibility supported by mixed-method evidence, rather than a definitive evaluation of impact.

The findings also need to be situated within a wider evidence base that includes mixed outcomes. While many studies report AR's capacity to increase engagement and motivation [47, 49], meta-analytic evidence indicates that AR can increase cognitive load under certain conditions, particularly when tasks are complex or when visual information becomes distracting [5]. Similarly, while GenAI scaffolds can support learning and organisation, concerns remain about learner dependency and reduced autonomy if scaffolding becomes overly directive or replaces sense-making processes [9]. These perspectives underline the importance of contextual sensitivity, clear pedagogical framing, and responsible design choices (e.g., minimalist UI, short prompts, and modular access) that reduce overload and preserve learner agency.

7.7 Implications for inclusive learning design and future research

This pilot DBR evaluation suggests that integrating AR and GenAI can offer complementary support for ADHD-related challenges in higher education, particularly in task salience, initiation, and perceived cognitive load. Crucially, the findings point to mechanisms rather than generic claims: AR appeared to support externalisation of tasks and deadlines in ways participants found harder to ignore, while GenAI provided just-in-time scaffolding that reduced uncertainty about how to start or how to break tasks into manageable components. For ADHD learners, support for “what do I do next?” may be as consequential as support for attention itself.

A key implication for inclusive learning design is that effectiveness may depend less on technological sophistication and more on interaction shape and cognitive accessibility. Participants valued simple interfaces, predictable workflows, and short actionable prompts, supporting a “minimum necessary complexity” principle for neurodiverse learners, where reducing friction and cognitive overhead is a primary design objective. The study also suggests AR and GenAI may be most beneficial when positioned as

scaffolds that enhance agency rather than substitutes for learning; participants described the most useful GenAI support as guidance that enabled them to proceed independently.

Future research should evaluate the approach in authentic course contexts over longer durations, with larger samples and stronger comparison conditions. This would enable examination of sustained outcomes, differential effects across ADHD presentations, and potential reliance risks (e.g., over-dependence on AI prompts). It would also allow investigation of practical implementation constraints, including staff workload, institutional policy alignment, and data governance considerations relevant to deploying GenAI-supported systems in higher education.

8 Conclusion and future work

As a pilot study with a modest sample, the findings primarily demonstrate feasibility and illuminate plausible mechanisms of support rather than providing generalisable evidence of effectiveness. We introduced ADHDvance LearnAR, a user-centred application that integrates Augmented Reality and Generative AI to support undergraduate students diagnosed with ADHD, and evaluated it through a Design-Based Research approach. Although quantitative differences across conditions did not reach statistical significance, the consistent directional trends, together with detailed qualitative accounts, suggest that AR may strengthen task salience and sustained engagement through externalisation, while GenAI may reduce planning friction and cognitive load through just-in-time scaffolding and simplification.

The study therefore contributes evidence that the value of these technologies for ADHD learners may lie in how they shape interaction and reduce executive demands: making tasks harder to dismiss, supporting initiation and re-entry, and lowering the effort required to interpret dense information. At the same time, the limited sample and short exposure period constrain inference and emphasise the need to treat the outcomes as exploratory.

Future work should prioritise longitudinal, larger-scale evaluations in authentic course contexts, using stronger comparison designs to examine sustained academic, wellbeing, and social outcomes. Further investigation is also warranted into which design elements drive benefit (e.g., adaptive feedback, multimodal representations, or carefully bounded gamification), alongside ethical and equity considerations for GenAI deployment in inclusive education. Overall, this work supports the development of human-centred, cognitively accessible educational technologies that can be tailored to the diverse needs of neurodiverse learners.

Author contributions

S.S.O conceptualized the study. R.P, S.S.O, & K.A designed the study, performed the user evaluation and methodology. R.P implemented the application and wrote the draft and visualized the figures. The final manuscript has been reviewed and edited by S.S.O. and K.A. All authors have read and agreed to the published version of the manuscript.

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Data availability

Data sets generated during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics and consent to participate

The authors of this manuscript confirms that: Informed consent was obtained from all the participants. All procedures followed were by the ethical standards of the responsible committee on human experimentation at the University of

Exeter, and with the Helsinki Declaration of 1964 and its later amendments. Ethical approval for the study was obtained from the Faculty of Environment, Science and Economy Research Ethics Committees at the University of Exeter.

Consent for publication

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Competing interests

The authors declare no competing interests.

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